

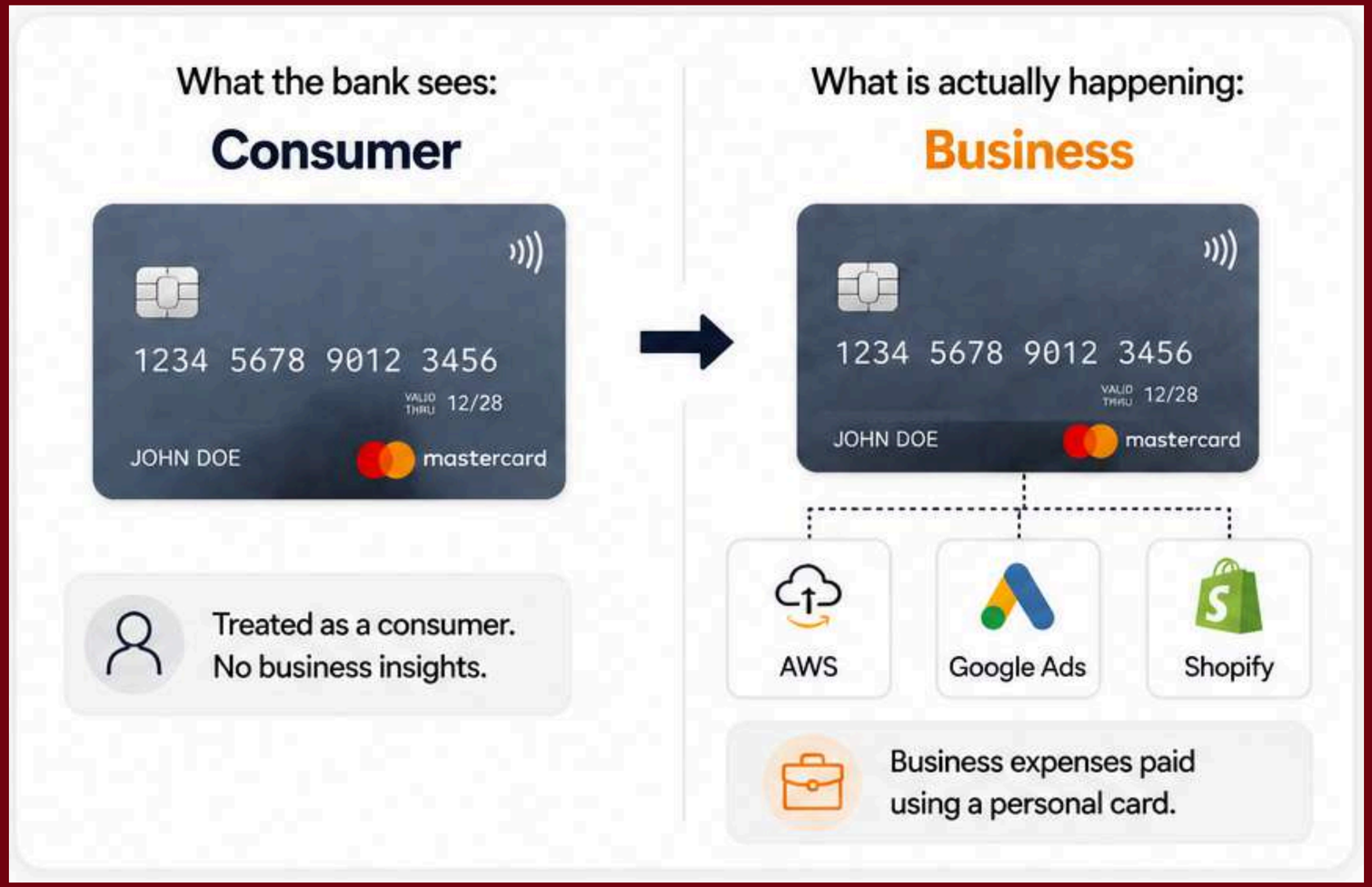


DETECTING HIDDEN ENTREPRENEURS

MASTERCARD DATAQUEST

PRESENTED BY : **TEAM ‘MIND MINERS’**

THE BANK SEES A CONSUMER CARD. THE BEHAVIOR SAYS BUSINESS.



HIDDEN BUSINESS SIGNALS:

- Inventory and wholesale purchases.
- Ads, software, and online tools.
- Recurring business payments.

WHY IT MATTERS:

- Missed business-card revenue.
- Missed cross-sell opportunities.
- Misclassfied commercial spending.

AGENDA: FROM HIDDEN TO DETECTED

PROBLEM	DATA	METHOD	RESULTS	ACTION
?	≈		✓	→
Hidden	**Measured**	**Corrected**	**Perfect**	**Ready**
Slide 4	Slides 5-7	Slides 8-11	Slides 12-13	Slide 14

EVERY HIDDEN ENTREPRENEUR IDENTIFIED IS A NEW BUSINESS BANKING CLIENT



HIGHER REVENUE PER CLIENT

Business cards carry higher transaction commissions.



CROSS-SELL POTENTIAL

These clients can be targeted with loans, POS terminals, payroll, cash registers.



BETTER DATA QUALITY

Commercial spending is currently miscounted as consumer behavior.

The bank already has their data. **The only missing piece is knowing who they are.**

3 DATASETS COVER 12.8 MILLION TRANSACTIONS ACROSS 105,000 CARDHOLDERS

BUSINESS CARDS

2,997,593

transactions

25,000 cards

CONSUMER CARDS

9,832,487

transactions

80,000 cards

MERCHANTS

2,165

unique merchants



BUSINESS CARDS BEHAVE DIFFERENTLY ACROSS EVERY DIMENSION WE MEASURED

Feature	Business	Consumer	Ratio
B2B merchant categories	61.4%	5.8%	10.6×
Recurring payments	15.5%	3.1%	4.9×
Average transaction (KZT)	167K	55K	3.1×
Online payments	84.7%	46.5%	1.8×
Weekend activity	12.5%	34.8%	0.36×

Note: 0.36× weekend activity = business cards avoid weekends. red = inverse signal (lower = more business-like)

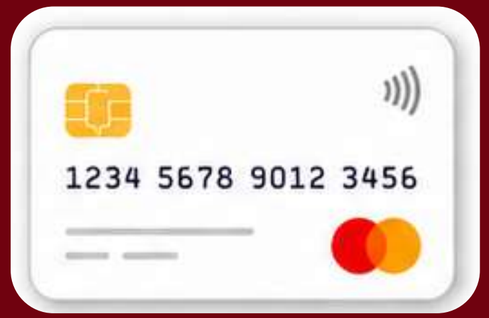
47 FEATURES, EACH BACKED BY A REAL BEHAVIORAL SIGNAL

1) CHANNEL
captures: How payments are made.
Online share: 84.8% vs 46.9%.

2) B2B SIGNALS
captures: What is being bought.
B2B MCC share: 10.6× higher.

3) AMOUNT
captures: Scale of spending.
Median: 6.5× higher in business.

4) TEMPORAL
captures: When the card operates.
Weekend share: 2.8× lower business.



RAW TRANSACTIONS

5) RECURRING
captures: Platform billing patterns.
Recurring share: 4.9× higher.

6) DIVERSITY
captures: Merchant concentration.
Unique merchants: 0.46× (inverse).

7) VOLATILITY
captures: Order size variation.
Amount IQR: 3.8× higher.

Note: Every count also has a matching ratio so a card with 50 transactions and a card with 500 are compared fairly on behavior, not volume.

CARD-LEVEL 80/20 SPLIT ENSURES THE MODEL IS TESTED ON COMPLETELY UNSEEN PEOPLE

24,999 OUT OF 25,000 BUSINESS CARDS ARE ACTIVE IN ALL 6-MONTHS.

This means almost every cardholder has a complete 6-month behavioral profile.
The right question is: can the model identify a hidden entrepreneur it has never encountered before?

APPROACH:

Features built from each card's full 6-month history → Cards split 80/20 by identity — no person appears in both sets. → Test set: 21,000 completely new cardholders the model never touched.



TRAIN
84,000
CARDS



TEST
21,000
CARDS

HIDDEN ENTREPRENEURS CONTAMINATE THE TRAINING DATA - PU LEARNING FINDS THEM FIRST

STEP 1 - THE PROBLEM

Standard approach
Label all 80,000 consumers as "not a business"

▼

Mislabaled Data
Some of those labels are WRONG - hidden businesses buried in consumer pool

▼

Corrupted Model
Model learns incorrect patterns from bad labels

STEP 2 - OUR SOLUTION

Inject 3,000 Spy Cards
Confirmed business cards planted into the consumer pool

▼

Score Every Consumer
Train spy detector → each consumer gets a probability score

▼

Apply Threshold = 0.923608
Consumers below threshold are Reliable Negatives - genuinely not businesses

▼

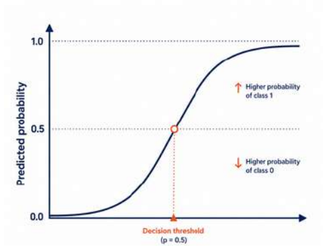
Train on Clean Data Only
Model learns from verified labels - no contamination

Source: Elkan, C. & Noto, K. (2008). Learning Classifiers from Only Positive and Unlabeled Data. KDD '08.

LIGHTGBM OUTPERFORMS THE BASELINE AND IS THE RIGHT TOOL FOR THIS PROBLEM

LOGISTIC REGRESSION

Baseline



- Simple linear boundary
- Requires feature scaling
- Good interpretability

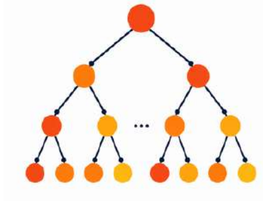
AUC: 1.0000

F1: 0.9997

LIGHTGBM

Best Model

Selected



- Gradient boosting, 187 trees
- Handles mixed features natively
- Built-in imbalance control

AUC: 1.0000

F1: 0.9999

Why LightGBM?

No feature scaling needed.

scale_pos_weight=3.20 handles 3.2:1 class imbalance directly.

Early stopping at 187 trees prevents overfitting.

Best F1 score on the held-out test set.

FIVE PARAMETER COMBINATIONS WERE TESTED AND THE BEST CONFIGURATION WAS SELECTED

Combo	**Max Leaves** (tree complexity)	**Learning Rate** (step size)	**Min Samples** (per leaf)	**Row Sample** (data used per tree)	**Col Sample** (features per tree)	**CV AUC** (accuracy score)
1 ✓	**31**	**0.1**	**10**	**0.7**	**0.7**	**1.0000**
2	63	0.05	20	0.8	0.8	1.0000
3	127	0.03	30	0.9	0.9	1.0000
4	50	0.07	15	0.6	0.8	1.0000
5	90	0.04	25	0.8	0.7	1.0000

THE MODEL FINDS 4,999 OUT OF 5,000 ENTREPRENEURS WITH ZERO FALSE ACCUSATIONS

1.0000	0.9999	1.0000	0.9998
AUC-ROC	F1 Score	Precision	Recall

	Predicted Consumer	Predicted Entrepreneur
Actual Consumer	**16,000** ✓True Negative	0 ✗False Positive
Actual Entrepreneur	0 ✗False Negative	**4,999** ✓True Positive

ALL 80,000 CONSUMER CARDS ARE RANKED AND THE TOP CARDS SHOW CLEAR BUSINESS PATTERNS

TOP SCORED CONSUMER CARDS

Cards	Score	Online %	B2B MCC %	Weekend %
1522*	0.402	70.6%	73.5%	20.6%
8610*	0.299	77.1%	65.7%	14.3%
1058*	0.076	94.0%	89.6%	14.3%

*Every consumer card recevies a **score**, and outreach can be scaled from 800 high-priority cards to 8,000 borad-coverage targets.*

OUTREACH OPTIONS

Threshold	Cards	Use Case
TOP 1%	800 cards	Relationship manager calls
TOP 5%	4,003 cards	Digital outreach campaign
TOP 10%	8,000 cards	Maximum coverage

FINALLY, THE BANKS CAN TURN MODEL SCORES INTO BUSINESS ACTIONS

01 **PHASE 01 IMMEDIATE**

Send the top 800 scored cards to business banking managers. Start with a simple offer: a business card with clearer pricing and benefits.

02 **PHASE 02 CROSS-SELL**

After confirming business activity, suggest products that match their needs: working capital loans, POS terminals, payroll tools,cash register

03 **PHASE 03 CALIBRATE**

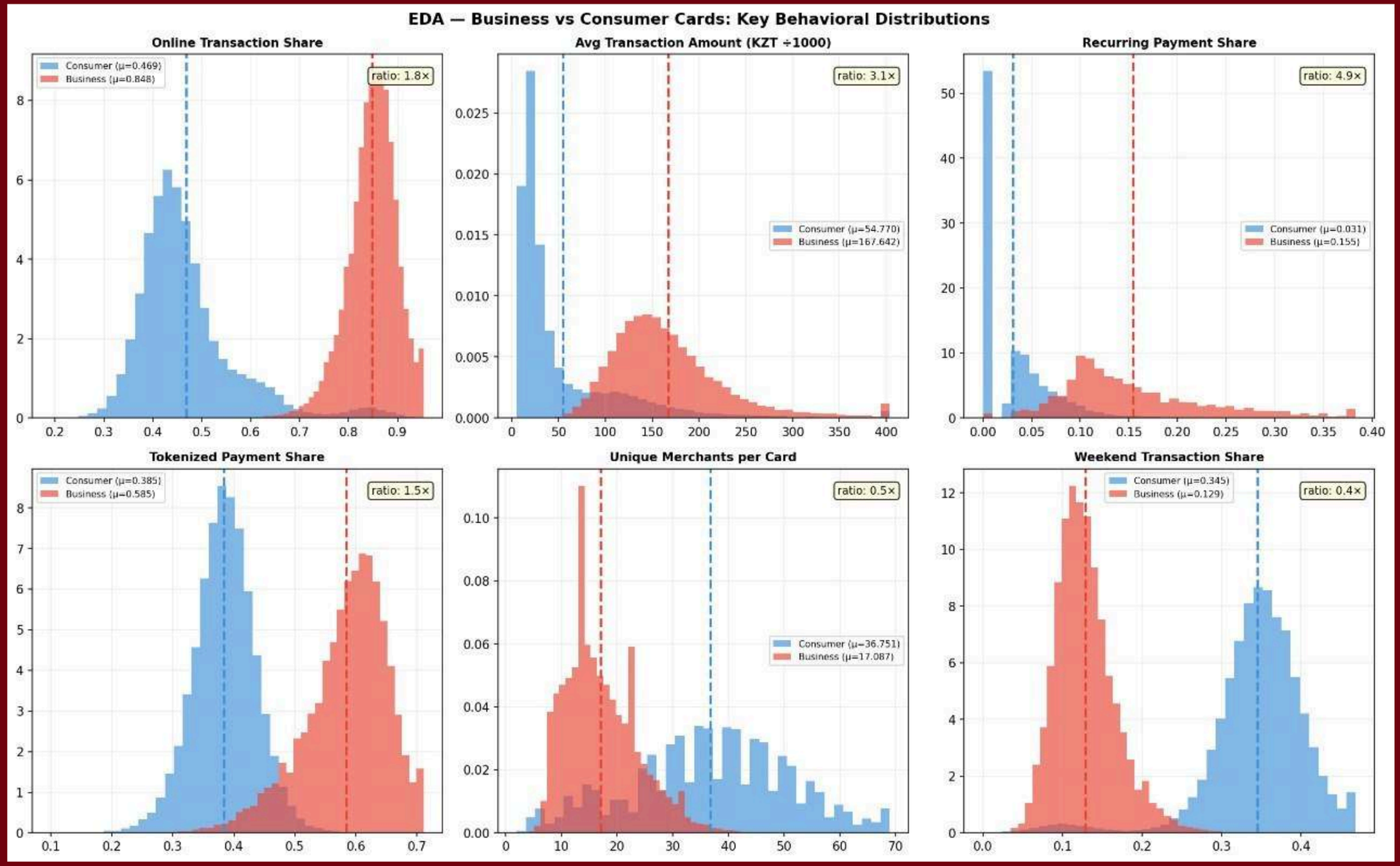
Adjust the score cutoff based on team capacity. A lower cutoff gives more leads; a higher cutoff gives fewer but stronger leads.

04 **PHASE 04 MAINTAIN**

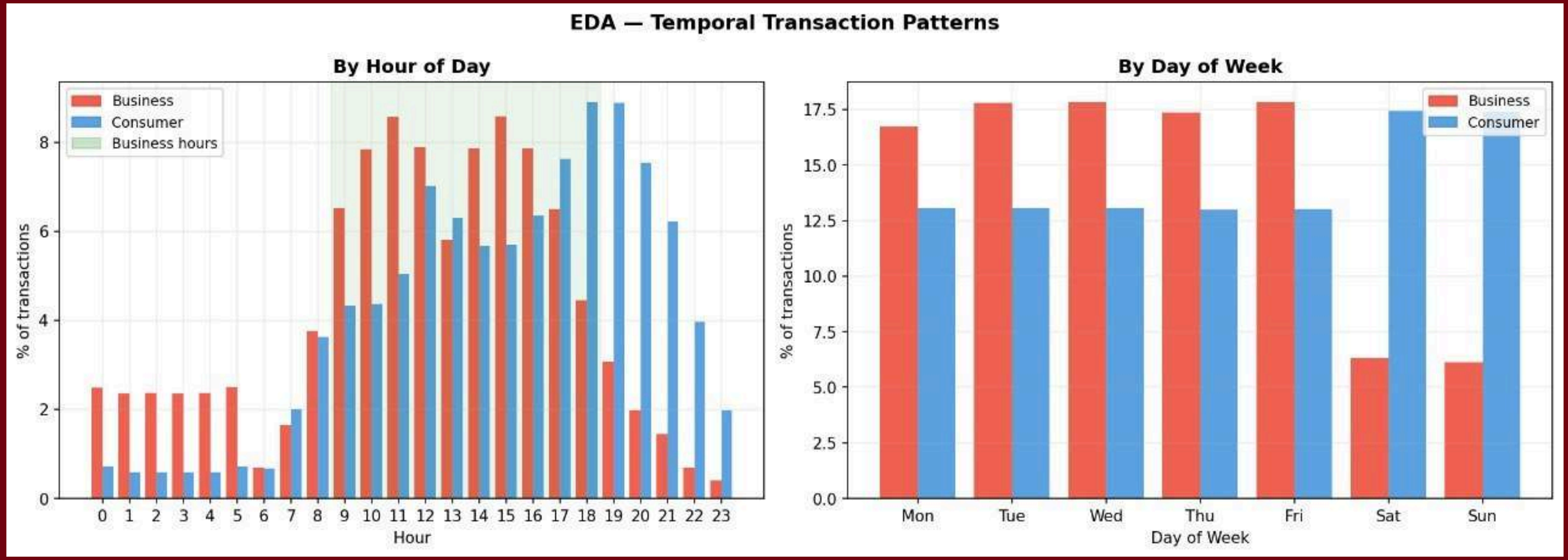
Retrain the model monthly, watch for changes in customer behavior, and update the B2B MCC list with help from banking and merchant experts.

Limitations to note: Synthetic AUC = 1.0 | Label noise in business cards | No temporal trend features | No network features

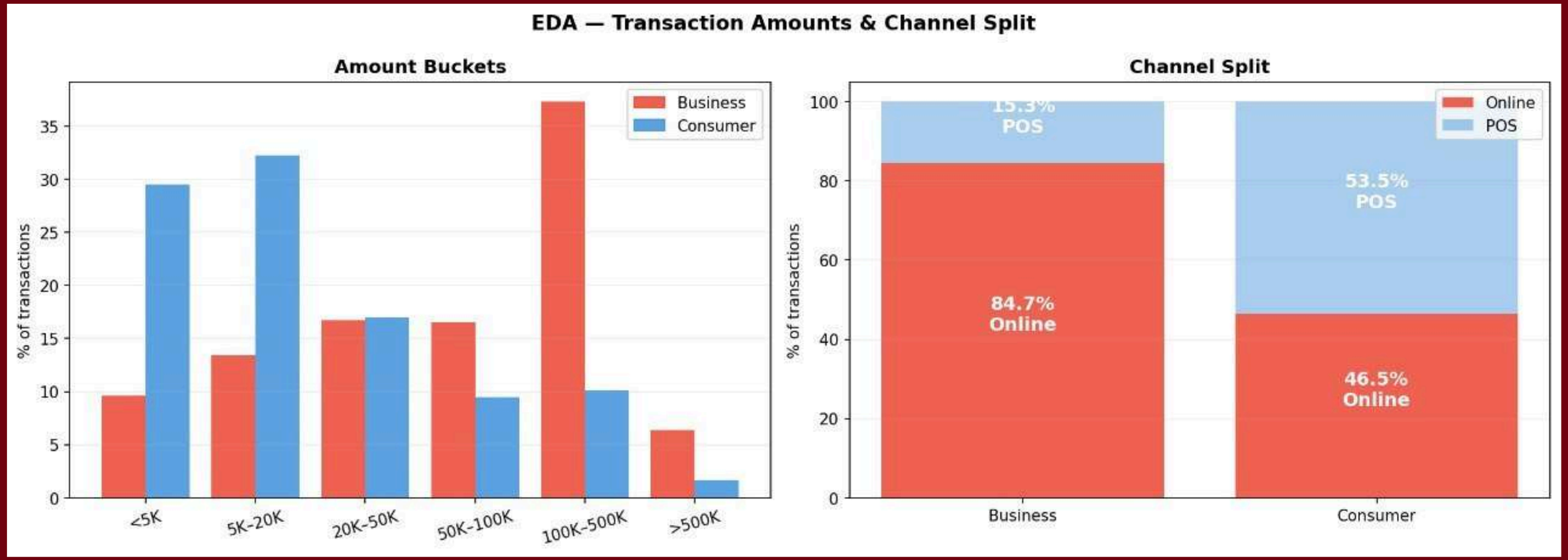
APPENDIX 1/7


EDA — Business vs Consumer Cards: Key Behavioral Distributions
Online Transaction Share
Consumer (μ=0.469)
Business (μ=0.848)
ratio: 1.8×
Avg Transaction Amount (KZT ÷1000)
Consumer (μ=54.770)
Business (μ=167.642)
ratio: 3.1×
Recurring Payment Share
Consumer (μ=0.031)
Business (μ=0.155)
ratio: 4.9×
Tokenized Payment Share
Consumer (μ=0.385)
Business (μ=0.585)
ratio: 1.5×
Unique Merchants per Card
Consumer (μ=36.751)
Business (μ=17.087)
ratio: 0.5×
Weekend Transaction Share
Consumer (μ=0.345)
Business (μ=0.129)
ratio: 0.4×

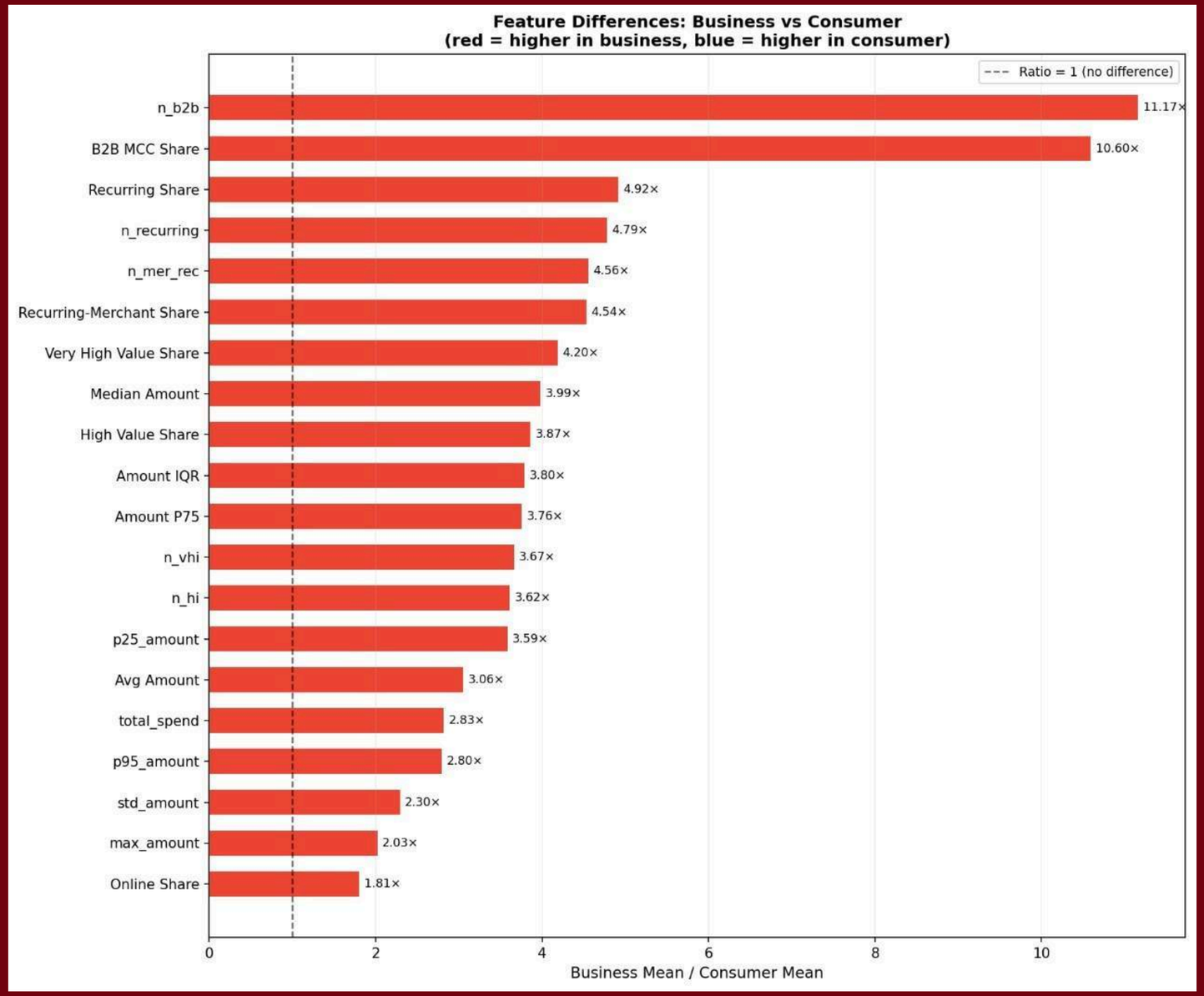
APPENDIX 2/7


EDA — Temporal Transaction Patterns
By Hour of Day
Business
Consumer
Business hours
% of transactions
Hour
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
By Day of Week
Business
Consumer
% of transactions
Day of Week
Mon Tue Wed Thu Fri Sat Sun

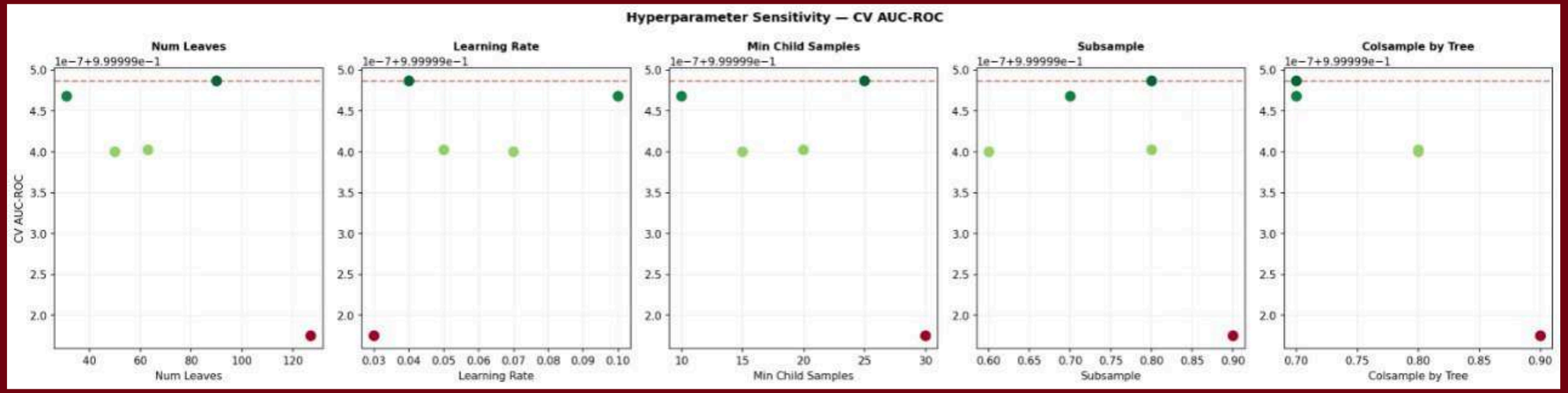
APPENDIX 3/7

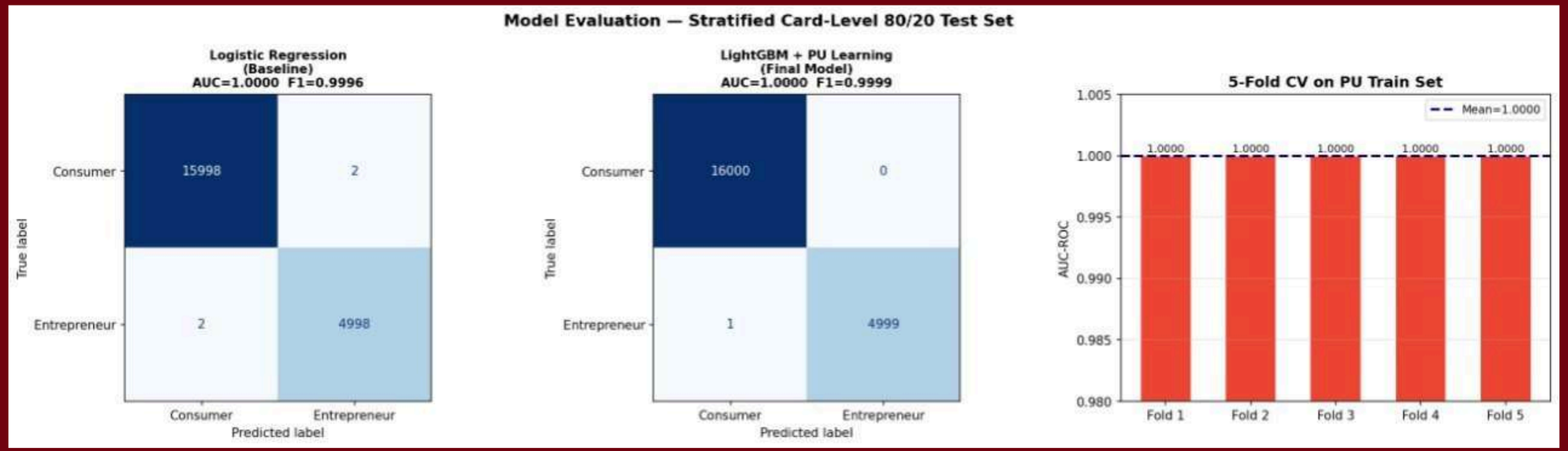

EDA — Transaction Amounts & Channel Split
Amount Buckets
% of transactions
Business
Consumer
<5K
5K-20K
20K-50K
50K-100K
100K-500K
>500K
Channel Split
% of transactions
Online
POS
15.3%
POS
84.7%
Online
53.5%
POS
46.5%
Online
Business
Consumer

APPENDIX 4/7

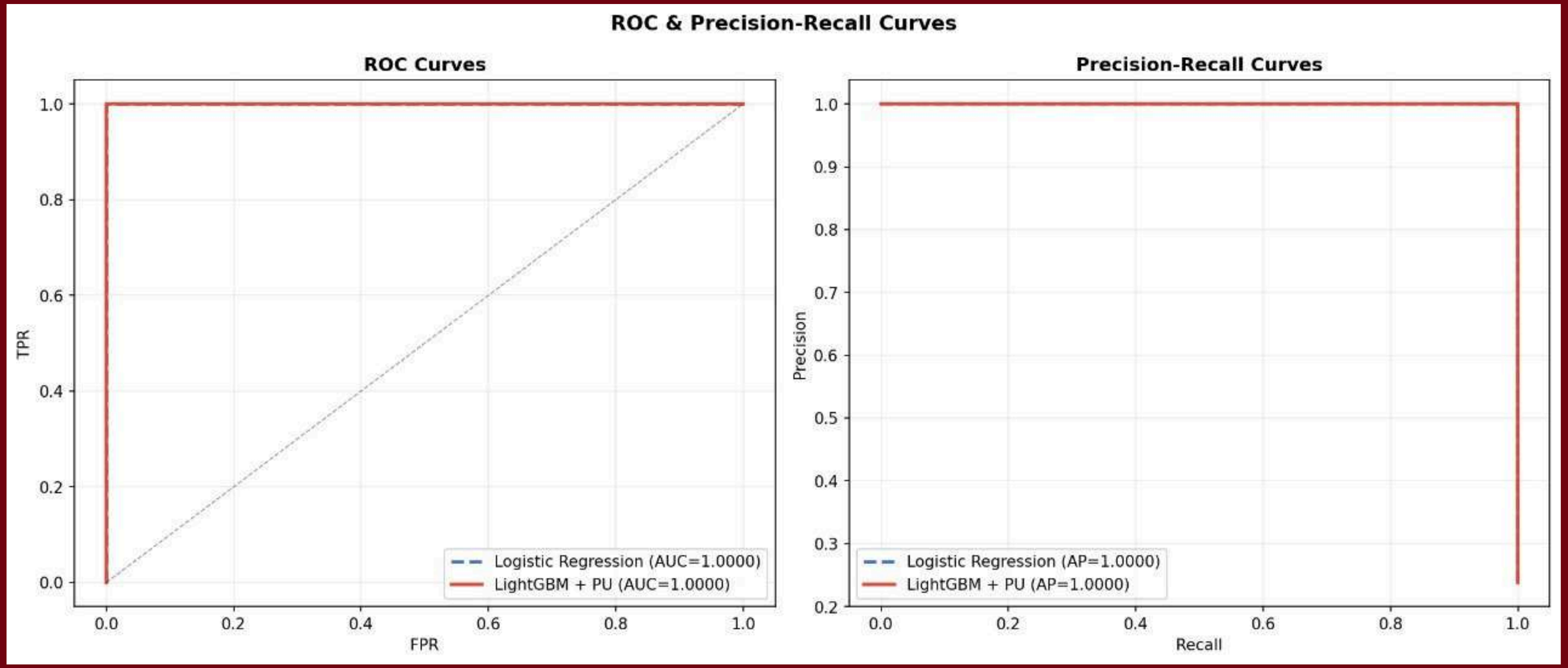

Feature Differences: Business vs Consumer
(red = higher in business, blue = higher in consumer)
Ratio = 1 (no difference)
n_b2b
B2B MCC Share
Recurring Share
n_recurring
n_mer_rec
Recurring-Merchant Share
Very High Value Share
Median Amount
High Value Share
Amount IQR
Amount P75
n_vhi
n_hi
p25_amount
Avg Amount
total_spend
p95_amount
std_amount
max_amount
Online Share
11.17×
10.60×
4.92×
4.79×
4.56×
4.54×
4.20×
3.99×
3.87×
3.80×
3.76×
3.67×
3.62×
3.59×
3.06×
2.83×
2.80×
2.30×
2.03×
1.81×
0
2
4
6
8
10
Business Mean / Consumer Mean

APPENDIX 5/7

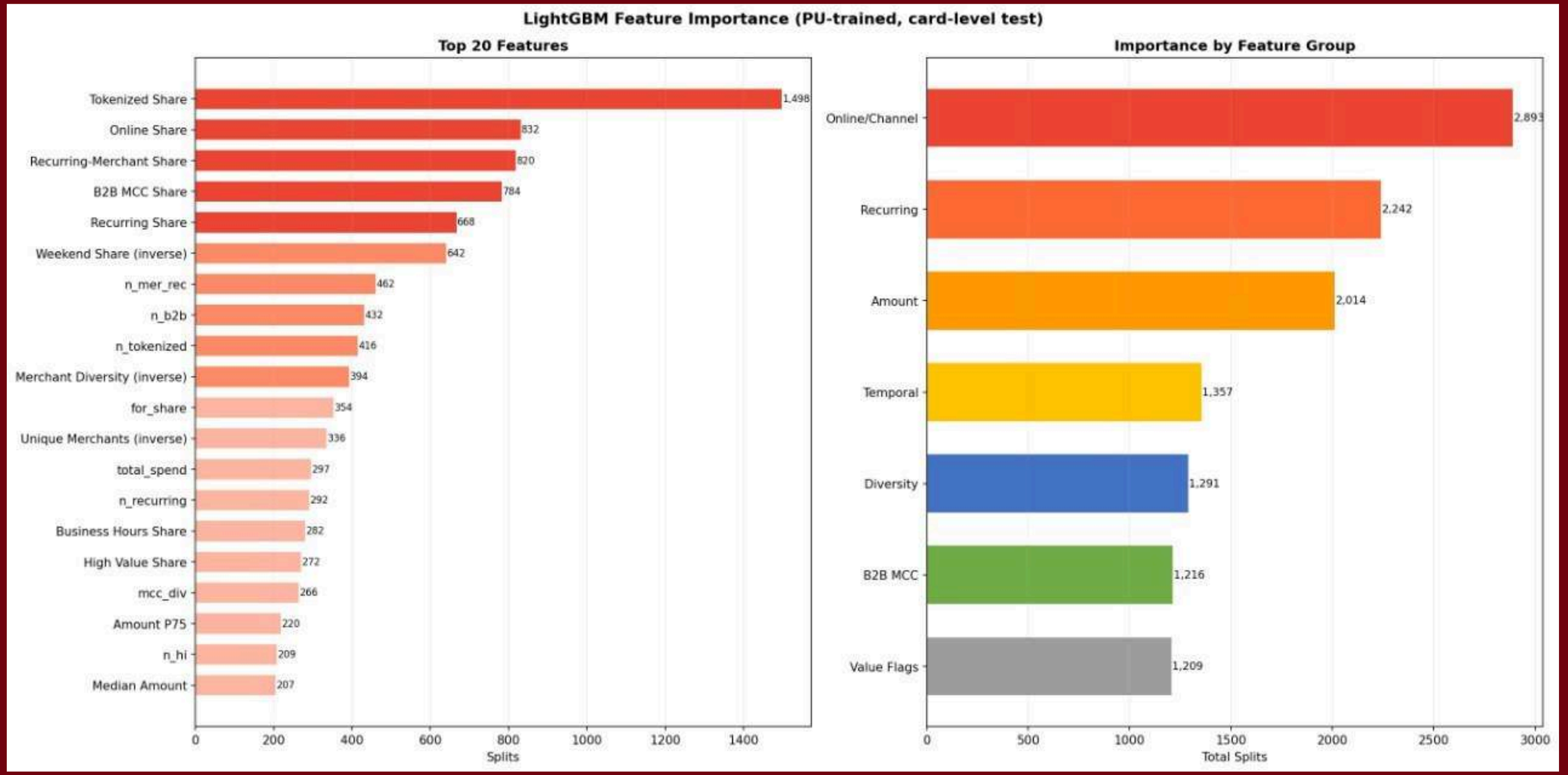

Hyperparameter Sensitivity — CV AUC-ROC
Num Leaves
Learning Rate
Min Child Samples
Subsample
Colsample by Tree
1e−7+9.99999e−1
CV AUC-ROC
Num Leaves
Learning Rate
Min Child Samples
Subsample
Colsample by Tree


Model Evaluation — Stratified Card-Level 80/20 Test Set
Logistic Regression
(Baseline)
AUC=1.0000 F1=0.9996
LightGBM + PU Learning
(Final Model)
AUC=1.0000 F1=0.9999
Consumer
Entrepreneur
True label
Predicted label
15998
2
2
4998
16000
0
1
4999
5-Fold CV on PU Train Set
Mean=1.0000
1.0000
AUC-ROC
Fold 1
Fold 2
Fold 3
Fold 4
Fold 5

APPENDIX 6/7


ROC & Precision-Recall Curves
ROC Curves
TPR
FPR
0.0
0.2
0.4
0.6
0.8
1.0
Logistic Regression (AUC=1.0000)
LightGBM + PU (AUC=1.0000)
Precision-Recall Curves
Precision
Recall
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Logistic Regression (AP=1.0000)
LightGBM + PU (AP=1.0000)

APPENDIX 7/7


LightGBM Feature Importance (PU-trained, card-level test)
Top 20 Features
Tokenized Share 1,498
Online Share 832
Recurring-Merchant Share 820
B2B MCC Share 784
Recurring Share 668
Weekend Share (inverse) 642
n_mer_rec 462
n_b2b 432
n_tokenized 416
Merchant Diversity (inverse) 394
for_share 354
Unique Merchants (inverse) 336
total_spend 297
n_recurring 292
Business Hours Share 282
High Value Share 272
mcc_div 266
Amount P75 220
n_hi 209
Median Amount 207
0 200 400 600 800 1000 1200 1400
Splits
Importance by Feature Group
Online/Channel 2,893
Recurring 2,242
Amount 2,014
Temporal 1,357
Diversity 1,291
B2B MCC 1,216
Value Flags 1,209
0 500 1000 1500 2000 2500 3000
Total Splits